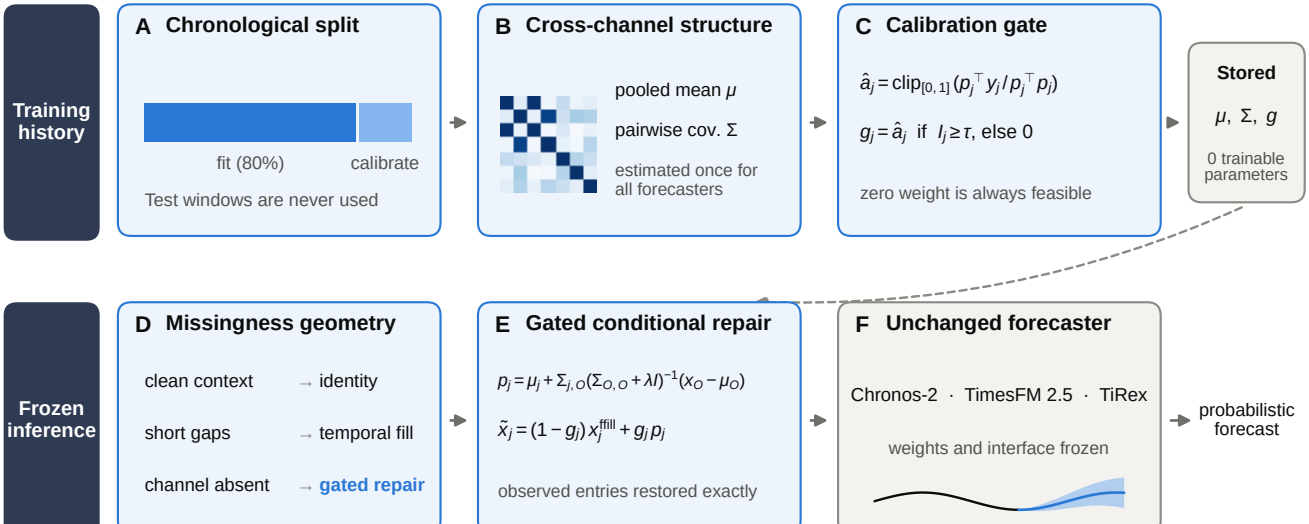
Training history
A Chronological split
fit (80%)
calibrate
Test windows are never used
B Cross-channel structure
pooled mean μ
pairwise cov. Σ
estimated once for all forecasters
C Calibration gate
$\hat{a}_j = \mathrm{clip}_{[0,1]}(p_j^\top y_j / p_j^\top p_j)$
$g_j = \hat{a}_j$ if $l_j \geq \tau$, else 0
zero weight is always feasible
Stored
μ, Σ, g
0 trainable parameters
Frozen inference
D Missingness geometry
clean context → identity
short gaps → temporal fill
channel absent → gated repair
E Gated conditional repair
$p_j = \mu_j + \Sigma_{j,O}(\Sigma_{O,O} + \lambda I)^{-1}(x_O - \mu_O)$
$\tilde{x}_j = (1 - g_j)\, x_j^{\mathrm{ffill}} + g_j\, p_j$
observed entries restored exactly
F Unchanged forecaster
Chronos-2 · TimesFM 2.5 · TiRex
weights and interface frozen
probabilistic forecast